

How Backtests Lie: Quantifying Lookahead Bias and Data Snooping

Martingale · Research Note 001 Author: hilothehappydog123-coder · Reproducible code: [experiment.py](#)

Abstract

A backtest that looks profitable is not evidence that a strategy works. Two methodological errors — **lookahead bias** and **data snooping** — routinely manufacture the appearance of alpha where none exists. We quantify both in controlled experiments where the ground-truth edge is known to be zero. We find that (1) a single-line, one-bar lookahead bug inflates a momentum strategy's annualized Sharpe ratio by **+1.12** (from 0.12 to 1.24), and (2) selecting the best of **337 strategies** on random-walk data yields an apparent in-sample Sharpe of **0.77** that collapses to **-0.54** out of sample. Both results are produced from a single reproducible script. The practical conclusion: a backtest is only as trustworthy as its methodology, and the two failures studied here are invisible in the equity curve.

1. Introduction

The first thing a new quant learns to do is backtest a strategy. The second thing — usually much later, and often expensively — is that most backtests are wrong. They are not wrong because the code has a crash; they are wrong because the *methodology* silently rewards the researcher for mistakes that would never survive live trading.

This note isolates and measures the two most common such mistakes. We deliberately use **simulated data**: to measure how much a bias distorts results, you need a world where the true edge is known. Real market data can never provide that, because you can never be sure whether a pattern is signal or luck. Simulation lets us set the true edge to exactly zero and watch the biases conjure profit from nothing.

2. Methodology

Data. All price paths are geometric Brownian motion generated by [quantsim](#), using the exact log-space solution so there is no discretization bias. Unless stated, paths use 20% annualized volatility over 4 years of daily bars.

The honest backtester. A strategy chooses a target weight $w(t)$ using only prices through bar t . That weight earns the return from bar t to $t+1$ — never the return of a bar the strategy has already seen. Transaction costs of 1 basis point are charged on turnover. This no-lookahead rule is the single most important line in any backtester.

Reproducibility. Every number in this note is emitted by [experiment.py](#) and stored in [results.json](#). All randomness is seeded. Running the script reproduces the figures and the tables

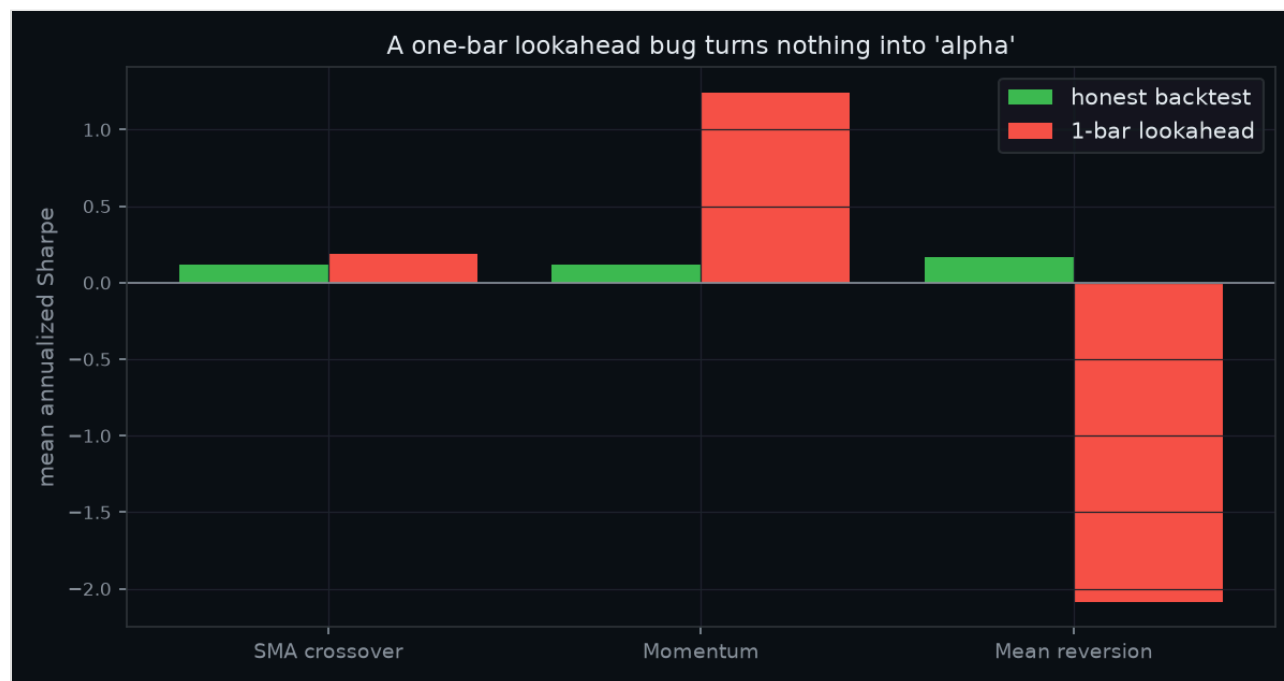
exactly.

3. Experiment 1 — Lookahead bias

Setup. We run three standard strategies (SMA crossover, momentum, mean reversion) on 500 independent price paths, twice each. The *honest* run uses the no-lookahead rule. The *cheat* run introduces a one-bar lookahead: the weight decided at bar t is instead credited the return of bar $t-1 \rightarrow t$ — the very bar the signal used to decide. This is the classic off-by-one that hides in a shifted index, and it is invisible in the output.

Results.

Strategy	Honest Sharpe	Lookahead Sharpe	Inflation
SMA crossover	+0.12	+0.19	+0.07
Momentum	+0.12	+1.24	+1.12
Mean reversion	+0.16	-2.08	-2.25



Interpretation. The one-bar lookahead does *not* inflate all strategies equally — and that is the important, non-obvious finding. Its effect depends on how the signal correlates with the bar it peeks at:

- **Momentum** buys *after* an up-move, so crediting it the up-move it just saw is enormously flattering: +1.12 Sharpe from a single shifted index. A strategy with no real edge (0.12) now looks genuinely good (1.24).
- **Mean reversion** buys *after* a down-move, so the same bug credits it the loss it saw, sending its Sharpe sharply *negative*. Lookahead corrupts; it does not merely inflate.

- **SMA crossover** is slow-moving, so a single bar barely changes the signal — the bug is nearly invisible here, which is exactly why it survives code review in trend systems.

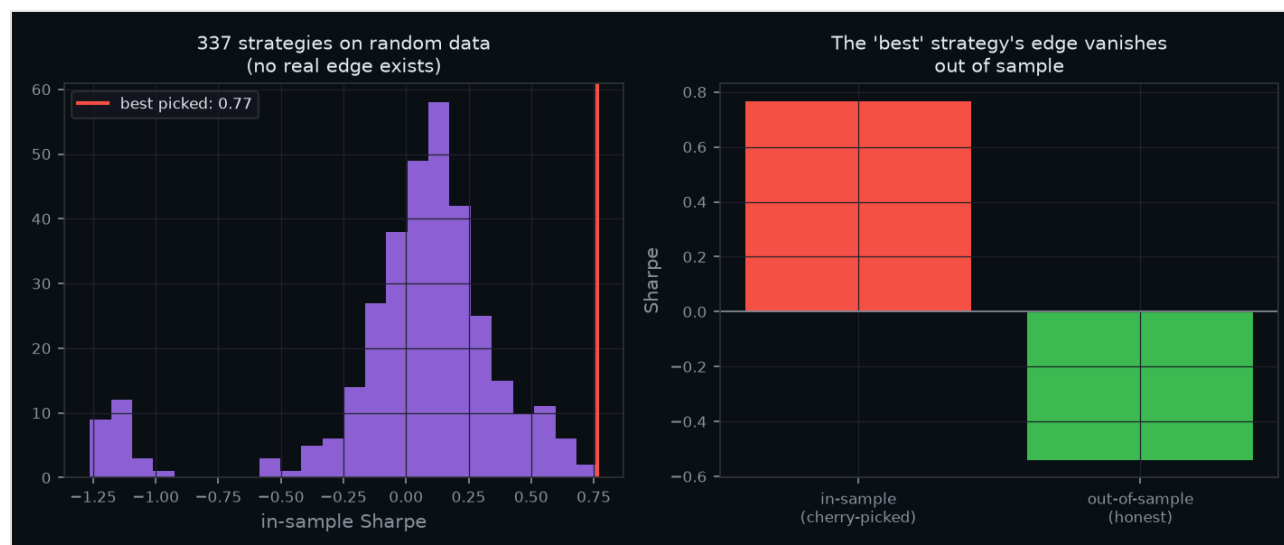
The danger is asymmetric: the bug most flatters the momentum-style strategies that beginners are most drawn to, in the direction (positive) most likely to be believed.

4. Experiment 2 — Data snooping

Setup. We generate a single **zero-drift** random-walk price series — a world with no edge, by construction. We then evaluate **337 strategies** (varied SMA, momentum, and mean-reversion parameters) on it, keep the one with the best in-sample Sharpe, and re-test that single winner on a fresh, independent random-walk series (out of sample).

Results.

Quantity	Sharpe
A single fixed strategy, averaged over fresh data	-0.12
Best of 337 strategies, in-sample (cherry-picked)	+0.77
That same "best" strategy, out-of-sample	-0.54



Interpretation. On data where profit is *impossible*, trying enough strategies and reporting the winner produces an in-sample Sharpe of 0.77 — a number most people would happily trade. It is pure survivorship of the luckiest parameter set. Out of sample the edge is not merely smaller; it is negative. The left histogram makes the mechanism visible: 337 draws from a noise distribution centered at zero, and we simply reported the right tail.

5. Discussion

Both experiments share one lesson: **the equity curve cannot tell you whether a backtest is honest**. A lookahead-inflated momentum curve and a cherry-picked random-data curve both look like discovered alpha. The only defenses are methodological and must be built in before results are ever seen:

1. **Enforce no-lookahead in code, and test for it.** The honest backtester above is worth more than any strategy run through it.
2. **Reserve out-of-sample data you never touch during search**, and judge a strategy only there.
3. **Count your attempts.** If you tried 300 strategies, a Sharpe of 0.77 is the *expected* best of 300 noise draws, not a discovery. Correct for multiple testing.
4. **Charge realistic costs**, which disproportionately punish the high-turnover strategies that in-sample search tends to select.

6. Limitations

This is a controlled simulation study, and its purpose is to measure biases, not to make claims about real markets. Real returns have fatter tails and volatility clustering that GBM omits; the *magnitude* of inflation on real data will differ. The lookahead form studied is one of several (others include using end-of-day data intraday, or survivorship in the asset universe). The data-snooping result uses a single random seed per sample; the qualitative conclusion is robust across seeds, but the exact Sharpe figures are seed-dependent. These are directions for Research Note 002.

7. Conclusion

A one-line bug turned a null momentum strategy into a Sharpe-1.24 "winner," and searching 337 strategies manufactured a 0.77 Sharpe from pure noise. Neither error is visible in the backtest output. The takeaway for anyone building trading strategies — including this author — is that the trustworthiness of a backtest lives entirely in its methodology, and both failures here are cheap to prevent and expensive to discover after the fact.

References

- Bailey, D., Borwein, J., López de Prado, M., & Zhu, Q. (2014). *Pseudo- Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting*.
- López de Prado, M. (2018). *Advances in Financial Machine Learning*.
- Harvey, C., Liu, Y., & Zhu, H. (2016). *...and the Cross-Section of Expected Returns*.

Reproduce: `pip install "git+https://github.com/hilothefunnydog123-coder/quantsim.git" matplotlib && python experiment.py`

How to cite

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